

Sales and Profitability Analysis Dashboard

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Role: Financial / Data Analyst

Tools Used: Power BI | Excel | DAX | Power Query | M language | R programming

Date Completed: 14/2/2026

Executive Summary

This project delivers a comprehensive business intelligence solution for Tailwind Traders, transforming raw transactional sales data into actionable insights to support strategic decision-making and enhance business performance. The objective was to prepare, model, and visualize sales and profitability data to enable clear understanding of revenue trends, product performance, and regional sales patterns.

The project began with data preparation and validation in Microsoft Excel, where sales transactions were structured, cleaned, and enriched with calculated metrics such as Gross Revenue, Tax, and Net Revenue. Following this, the dataset was integrated into Microsoft Power BI, where a robust snowflake schema was designed. User-created tables—including the Calendar Table, Exchange Data Table, and Sales in USD Table—were added, relationships defined, and DAX measures such as Gross Profit USD, Profit Margin %, Profit YTD %, Quarterly Profit, and Median Sales were implemented.

Two interactive reports—the Sales Report and the Profit Overview Report—were created, along with a combined dashboard featuring key KPIs and visualizations. The dashboards provide dynamic filtering, clear trend analysis, and actionable insights. Key findings include: the UK had the highest loyalty points, the UAE had the greatest median sales, Gardening products were the top-selling category, and the Modular Sofa Set generated the highest revenue.

This solution provides a scalable, accurate, and insightful reporting framework, enabling stakeholders to monitor performance, evaluate profitability, optimize inventory and marketing strategies, and make informed business decisions.

Business Problem

Tailwind Traders generates a high volume of daily sales transactions, but the data was previously scattered, inconsistent, and difficult to analyze, limiting the company's ability to track performance and identify trends. Critical challenges included:

- Difficulty in monitoring daily, monthly, and product-level sales.
- Inability to identify high- and low-performing products and regions.
- Risk of reporting errors due to inconsistent or missing data.
- Limited access to interactive insights for strategic decision-making.

The project aimed to address these issues by preparing clean data, developing a structured data model, and building interactive dashboards to provide clear visibility into sales performance, profitability, and customer engagement metrics.

Project Objectives

The objectives of this project were:

1. Prepare and Validate Data – Ensure accuracy, consistency, and readiness for analysis by cleaning, profiling, and standardizing sales datasets.
2. Calculate Key Financial Metrics – Compute Gross Revenue, Tax, Net Revenue, and other measures to support performance evaluation.
3. Develop a Data Model – Create relationships, user-defined tables, and DAX measures in Power BI for flexible, multi-dimensional analysis.
4. Create Interactive Visualizations – Develop dashboards that present sales, profitability, and customer engagement insights clearly and dynamically.
5. Support Data-Driven Decisions – Enable management to monitor trends, identify opportunities, and make strategic decisions with confidence.

These objectives ensured a full end-to-end business intelligence workflow from data preparation to actionable insights.

Data Source & Description

The project utilized multiple datasets from Tailwind Traders, provided in spreadsheet format. Together, these tables capture transactional sales, country-level mapping, and purchase records, forming a comprehensive view of business operations.

1. Tailwind Traders Sales

This dataset records daily sales transactions across products and regions.

Key Features:

- OrderID, CustomerID, Customer Name, Address, Contact Info (Phone, Email, Zip Code) – Identifies and profiles customers.
- Product Name, Product Description, Product Category, SKU Number, Attributes (Weight, Color, Size, Rating) – Enables product-level performance and attribute-based analysis.
- Quantity Purchased, Gross Product Price, Tax Per Product, Gross Revenue, Tax Paid, Net Revenue, Total Tax – Provides detailed financial metrics for revenue and tax analysis.
- Sales Rep, Loyalty Points, Stock – Supports sales team performance tracking and customer loyalty evaluation.
- Country ID – Links transactions to geographic regions via the Countries table.

This dataset covers all transactions during the reporting period, enabling trend identification, product performance assessment, and regional comparisons. Data cleaning addressed missing values, placeholder columns (e.g., "...26" to "...30"), and validation errors to ensure accuracy.

2. Countries

This reference table maps Country ID values to country names and exchange identifiers.

Key Features:

- Country ID – Primary key linking to the Sales dataset.
- Country – Provides geographic context (e.g., USA, UK, France, UAE, Australia).
- Exchange ID – Supports currency or market-specific analysis.

This table ensures consistency in regional reporting and enables cross-country comparisons.

3. Purchases

This dataset captures supplier-related purchase records, warranty details, and customer feedback.

Key Features:

- PurchaseID, Supplier, OrderID – Identifies purchase transactions and links them to sales orders.
- Purchase Date, Last Visited – Provides temporal context for supplier interactions.
- Returnstatus, Return Policy (Days), Warranty (Months) – Tracks product returns and warranty coverage.
- Feedback – Captures qualitative customer insights.

With 54 rows and 9 columns, this dataset complements the sales data by providing supplier-side information and customer satisfaction indicators.

Dataset Scope

- Comprehensive Coverage: Together, the three datasets provide a full view of sales, geography, and supplier interactions.
- Data Integrity: Cleaning and validation ensured consistency across tables, enabling reliable analysis.
- Analytical Potential: The combined structure supports time-based trend analysis, product performance evaluation, regional comparisons, and supplier/customer relationship insights.

Data Cleaning & Transformation

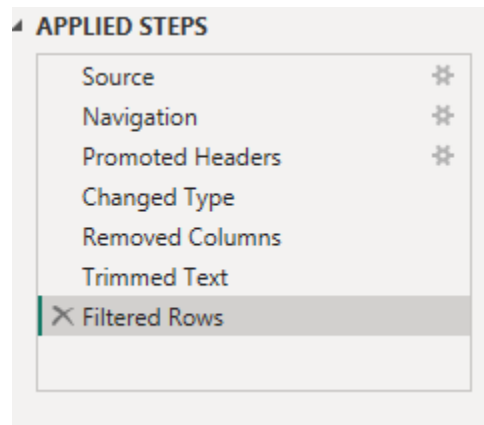
To ensure accurate and reliable reporting, a structured data preparation workflow was applied using Power Query and Excel, followed by modeling enhancements in Power BI.

1. Data Profiling and Validation

- Identified and addressed errors, empty cells, duplicates, and outliers.
- Reviewed statistical summaries to detect inconsistencies across sales, purchases, and country mappings.

2. Data Cleaning and Standardization

- Removed duplicate records to prevent double-counting.
- Trimmed spaces and standardized column headers for consistency.
- Corrected data types (numeric, categorical, date) to ensure accurate calculations and aggregations.



3. Feature Enrichment

- Added calculated fields such as Gross Revenue and Tax Paid.
- Integrated Purchases and Countries datasets to enable cross-functional analysis.

4. Data Modeling Enhancements

- Created a Calendar Table (DAX) to support time intelligence functions (YTD, QTD, seasonal trends).

```
1 CalendarTable =  
2 ADDCOLUMNS(  
3 CALENDAR(DATE(2020, 1, 1), DATE(2023, 12, 31)),  
4 "Year", YEAR([Date]),  
5 "Month Number", MONTH([Date]),  
6 "Month", FORMAT([Date], "MMMM"),  
7 "Quarter", QUARTER([Date]),  
8 "Weekday", WEEKDAY([Date]),  
9 "Day", DAY([Date])  
10 )
```

- Built an Exchange Data Table for currency conversion and international reporting.
- Developed a Sales in USD Table with advanced measures, including:
 - Gross Profit (USD)
 - Profit Margin %
 - Profit YTD %
 - Quarterly Profit
 - Median Sales

```

1 Sales in USD =
2 ADDCOLUMNS(
3     Sales,
4     "Country Name", RELATED(Countries[Country]),
5     "ExchangeRate", RELATED('ExchangeData'[ExchangeRate]),
6     "ExchangeCurrency", RELATED('ExchangeData'[Exchange Currency]),
7     "Gross Revenue USD", [Gross Revenue] * RELATED('ExchangeData'[ExchangeRate]),
8     "Net Revenue USD", [Net Revenue] * RELATED('ExchangeData'[ExchangeRate]),
9     "Total Tax USD", [Tax Paid] * RELATED('ExchangeData'[ExchangeRate])
10 )

```

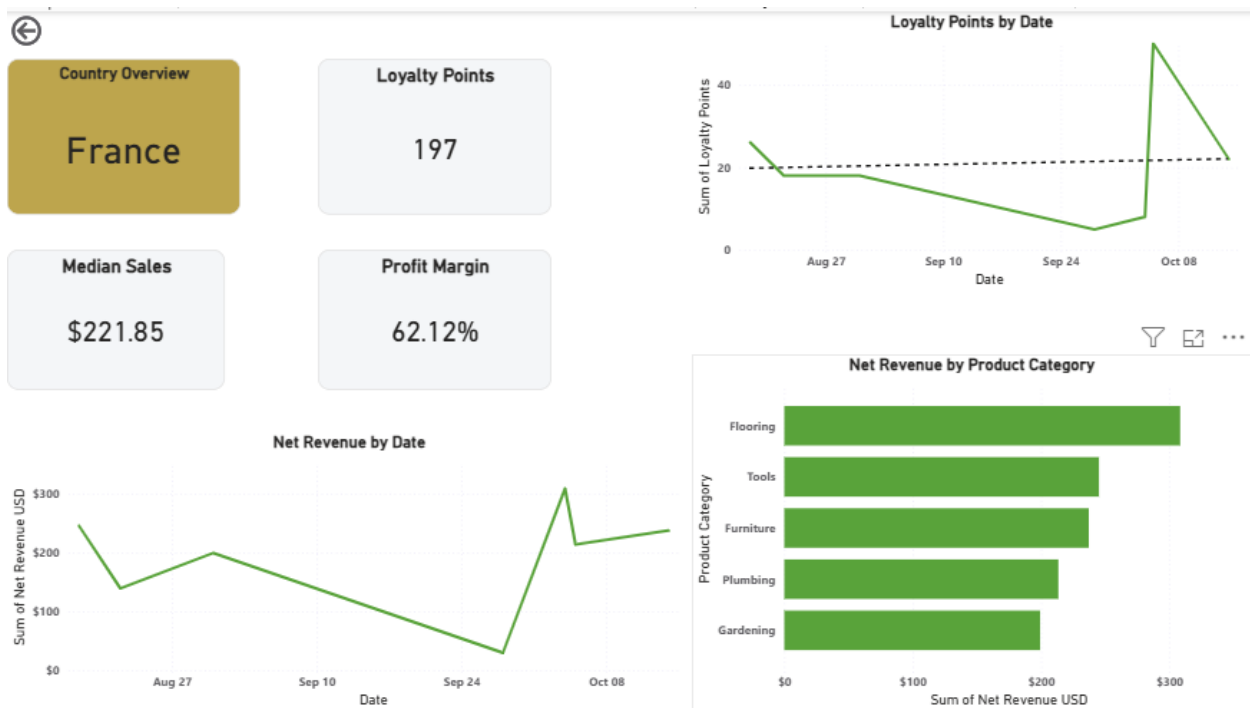
5. Drill-Through Functionality

To enhance interactivity and provide country-specific insights, a drill-through page was implemented with the following KPIs and visuals:

- Country Overview – High-level summary of performance.
- Median Sales – Central tendency of sales distribution.
- Loyalty Points – Customer engagement metric.
- Profit Margin – Profitability indicator.

Visuals included:

- Net Revenue by Date
- Net Revenue by Product Category
- Loyalty Points by Date



Data Modeling

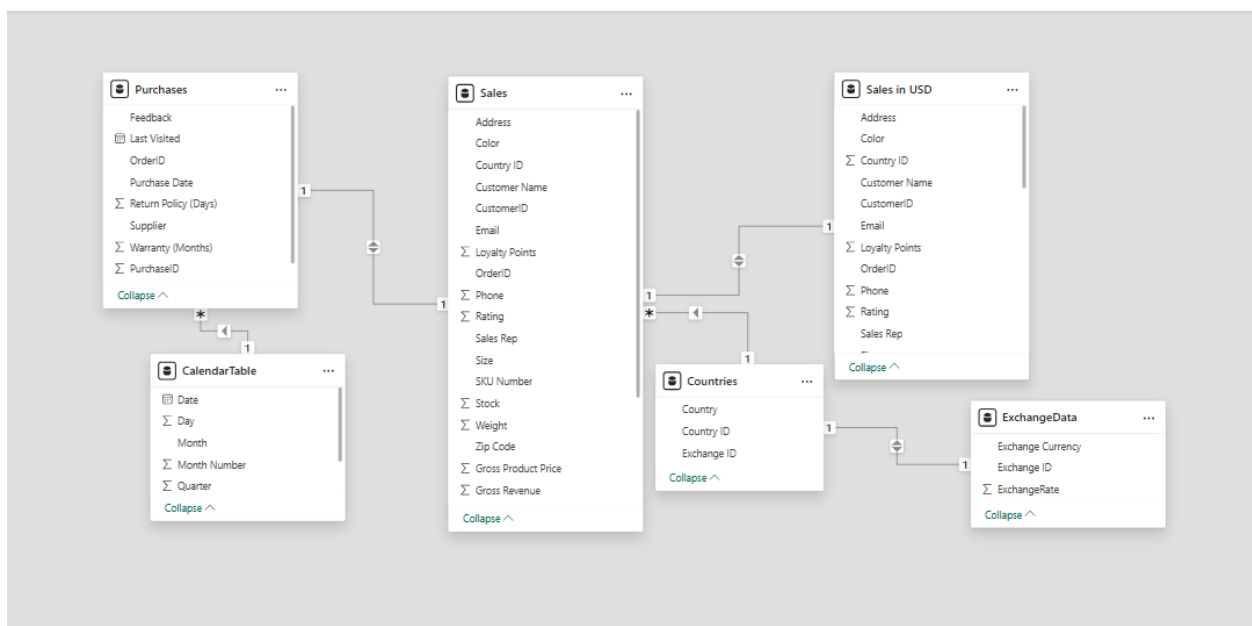
A snowflake schema was designed to structure the analytical model, ensuring scalability, clarity, and efficient querying across multiple dimensions of the Tailwind Traders datasets.

User-Created Tables

- Calendar Table – Provides a temporal dimension for time-based analysis, enabling quarterly, yearly, and seasonal performance metrics.
- Exchange Data Table – Stores exchange rates to support currency conversions and international comparisons.
- Sales in USD Table – Consolidates sales metrics into a single currency, facilitating advanced financial calculations and profitability analysis.

Key Relationships

- Sales ↔ Purchases – One-to-one relationship with bidirectional cross-filtering, linking customer transactions to supplier records.
- Sales ↔ Sales in USD – One-to-one relationship with bidirectional cross-filtering, ensuring consistent financial reporting across currencies.
- Purchases ↔ Calendar Table – Many-to-one relationship with single-direction cross-filtering, aligning supplier activity with time-based reporting.
- Sales ↔ Countries – Many-to-one relationship with single-direction cross-filtering, mapping transactions to geographic regions.
- Countries ↔ Exchange Data Table – One-to-one relationship with bidirectional cross-filtering, connecting regional sales to relevant exchange rates.



Analytical Capabilities

This model enables dynamic, multi-dimensional analysis across several domains:

- Sales Performance – Tracking gross, net, and tax-adjusted revenues.
- Profitability – Evaluating margins across products, regions, and currencies.
- Customer Engagement – Assessing loyalty points, feedback, and return behavior.
- Regional Insights – Comparing performance across countries with exchange rate normalization.
- Supplier Relations – Monitoring purchase activity, warranties, and return policies.

By integrating transactional, geographic, and supplier data into a unified schema, the model supports robust reporting, trend identification, and strategic decision-making.

Visualization & Dashboard Development

Sales Report

KPIs: Stock Levels, Quantity Purchased, Median Sales

Visuals:

- Loyalty Points by Country (Bar Chart)
- Median Sales Distribution by Country (Pie Chart)
- Median Sales Over Time (Line Chart)
- Quantity Sold by Product (Column Chart)

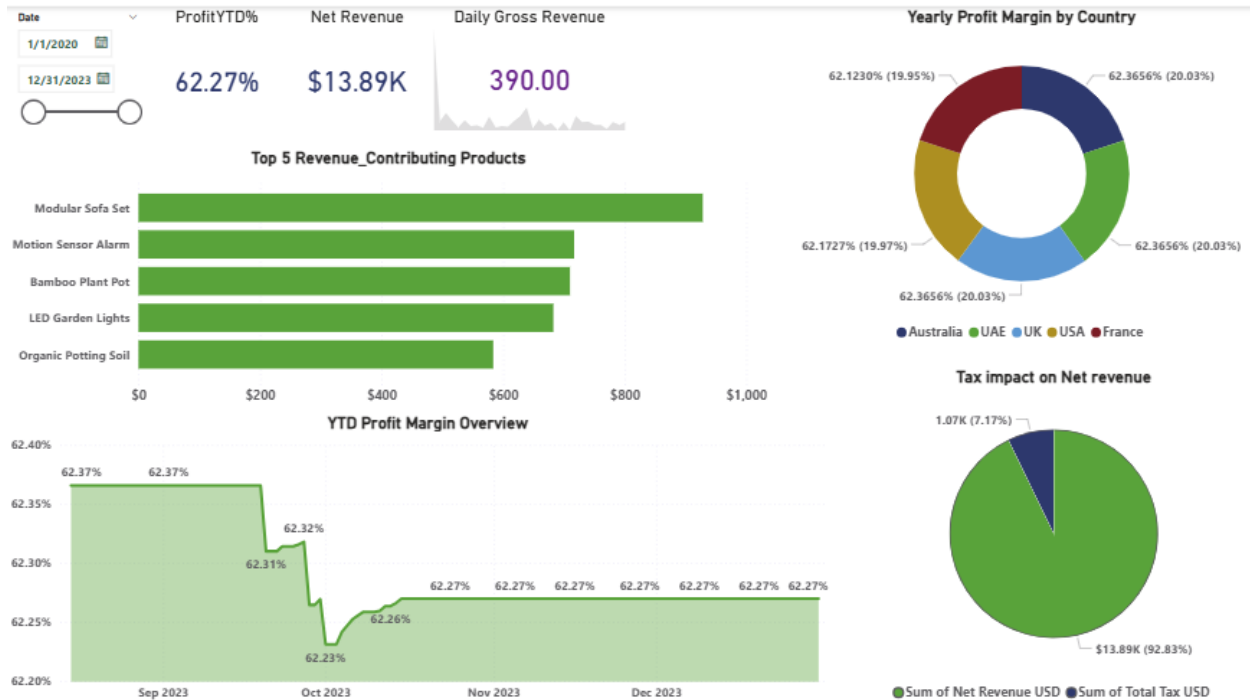


Profit Overview Report

KPIs: Profit YTD %, Net Revenue, Gross Revenue Over Time

Visuals:

- Profit Margin Over Time (Area Chart)
- Profit Margin by Country (Donut Chart)
- Net Revenue by Product (Column Chart)



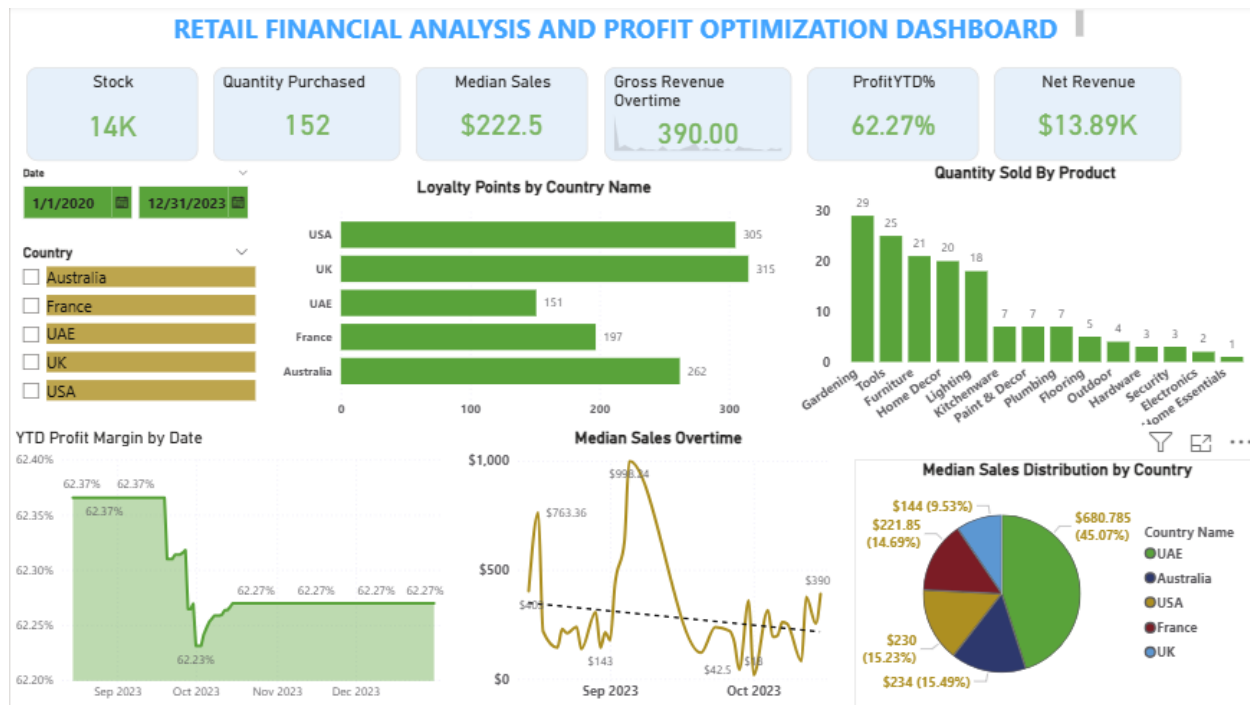
Combined Dashboard

KPIs:

- Sales: Stock, Quantity Purchased, Median Sales
- Profit: Profit YTD %, Net Revenue, Gross Revenue Over Time

Visuals:

- Profit Margin Over Time (Area Chart)
- Loyalty Points by Country (Bar Chart)
- Median Sales Over Time (Line Chart)
- Median Sales Distribution by Country (Pie Chart)
- Quantity Sold by Product (Column Chart)



This interactive dashboard unifies sales and profit metrics, allowing executives to monitor trends, compare regions and products, and make data-driven decisions efficiently.

Key Insights

1. Revenue & Profitability

- Net Revenue: \$13.89K with a ProfitYTD% around 62%, showing healthy profitability.
- Tax Impact: Taxes account for ~7% of net revenue, which is manageable but worth monitoring for optimization.
- Profit Margin Stability: Margins are very consistent (62.2–62.3%) across months and countries, suggesting strong cost control and pricing discipline.

2. Country Performance

- USA leads with ~45% of median sales, making it the largest market.
- UK and France contribute ~15% each, showing balanced mid-tier performance.
- Australia sits at ~14.7%, steady but not dominant.
- UAE lags at ~9.5%, both in sales and loyalty points, signaling weaker market penetration.
- Loyalty Points: UK (315) and USA (305) lead, while UAE (151) trails significantly.

3. Product Categories

- Top Sellers: Gardening (29), Tools (25), Furniture (21), Home Decor (20), Lighting (18).
- Moderate Sellers: Kitchen, Vehicle, Plumbing, Flooring.
- Weak Sellers: Electronics, Security, Home Essentials, Miscellaneous (≤ 3 units each).

- High Revenue Drivers: Modular Sofa Set, Motion Sensor Alarm, Bamboo Plant Pot, LED Garden Lights, Organic Potting Soil.

4. Sales Trends

- Median Sales Over Time: Highly volatile, with peaks near \$998 and troughs below \$100. This indicates seasonal demand spikes or promotional effects.
- Profit Margins: Despite sales volatility, margins remain stable—suggesting cost structures are resilient.

Recommendations

1. Country Strategy

- USA: Continue aggressive expansion—introduce premium product lines and loyalty-driven upselling.
- UK & France: Leverage strong loyalty to push higher-value products and bundles.
- Australia: Explore marketing campaigns to lift sales closer to USA levels.
- UAE: Needs targeted investment—localized promotions, partnerships, and culturally tailored product offerings.

2. Product Strategy

- Double Down on Winners: Expand Gardening, Furniture, and Home Decor lines with premium and eco-friendly variants.
- Revitalize Mid-tier Categories: Kitchenware and Vehicle products could benefit from bundled offers or seasonal promotions.
- Phase Out or Reassess Weak Categories: Electronics, Security, and Miscellaneous items may not justify inventory costs.

3. Customer Loyalty Programs

- Enhance loyalty rewards in weaker regions (UAE, France).
- Introduce tiered loyalty benefits (e.g., referral bonuses, exclusive discounts) to convert loyalty points into repeat purchases.

4. Revenue & Tax Optimization

- Explore tax-efficient strategies (e.g., regional warehousing, supplier negotiations).
- Monitor net revenue growth relative to tax impact to maintain profitability.

5. Sales Volatility Management

- Investigate causes of sharp sales peaks and troughs:
 - Seasonal demand?
 - Promotional campaigns?
 - Supply chain constraints?

Recommendation: Use predictive analytics to forecast demand and adjust inventory accordingly.

Technical Architecture Overview

The solution follows a structured business intelligence architecture:

Data Sources

→ Excel Transactional Data (Sales, Purchase, Countries)

Data Preparation

→ Power Query

- Removed duplicates
- Data profiling
- Standardized headers
- Trimmed data
- Corrected data types
- Handled missing values

Data Modeling

→ Snowflake Schema

- Sales
- Purchase
- Countries
- CalendarTable (Created using DAX)
- ExchangeData (Created)
- Sales in USD (Created)

Relationships included:

- One-to-one with both cross-filter (Sales ↔ Purchase)
- Many-to-one with single cross-filter (Sales ↔ Countries)
- Time intelligence via CalendarTable
- Currency conversion via ExchangeData

Analytics Layer

→ DAX Measures

- Gross Profit USD
- Profit Margin %
- Profit YTD %
- Quarterly Profit
- Median Sales

Presentation Layer

→ Two Reports + Consolidated Dashboard

This layered approach ensures scalability, performance optimization, and analytical flexibility.

Dashboard Interactivity & Analytical Capabilities

The dashboards were designed to allow dynamic exploration of data through:

- Cross-filtering between visuals
- Time-based analysis via CalendarTable (YTD %, Quarterly trends)
- Product-level drill-down
- Country-level performance comparison
- Real-time currency conversion through Sales in USD

Users can analyze:

- Revenue trends over time
- Profit margin fluctuations
- Regional contribution to sales
- Product-level revenue and volume

This interactivity enables deeper analytical insights beyond static reporting.

Challenges and Solutions

1. Missing Cost Data

Challenge: Cost per unit was not available in the transactional dataset, which limited direct profit margin calculations.

Solution: To address this, cost data was estimated using a structured assumption:

Cost per Unit = Gross Product Price × 0.35

This approach provided a consistent baseline for margin analysis while maintaining transparency in reporting.

Future Improvement: Integrate actual purchase cost data from supplier records to enable precise gross margin calculations and reduce reliance on assumptions.

2. Inconsistent Data & Missing Values

Challenge:

- Empty cells
- Outliers
- Inconsistent headers
- Duplicates

Solution:

- Conducted data profiling in Power Query.
- Removed duplicate records to prevent double-counting.
- Standardized headers and trimmed spaces.
- Validated data integrity through statistical summaries.

3. Multi-Currency Reporting

Challenge: Sales occurred across multiple regions, requiring currency standardization for consolidated reporting.

Solution:

- Created an Exchange Data Table to store exchange rates.
- Built a Sales in USD calculated table for unified financial reporting.
- Implemented DAX measures to ensure consistency across profitability metrics.

Outcome

By addressing these challenges, the project achieved a clean, standardized, and analytically robust dataset. The cost estimation method allowed for interim profitability insights, while the multi-currency framework and data validation ensured reliable, comparable reporting across regions.

Limitations & Future Enhancements

- Margin calculations were based on estimated cost assumptions.
- Real-time exchange rate integration via API could improve accuracy.
- Row-Level Security (RLS) could be implemented for regional managers.
- Integration with ERP systems would enable automated data refresh.
- Predictive forecasting using historical sales data could enhance strategic planning.

Skills Demonstrated

This project highlights the following competencies:

Data Preparation:

- Power Query profiling
- Data cleaning and transformation
- Standardization and validation

Data Modeling:

- Snowflake schema design
- Relationship optimization
- Cross-filter direction management

DAX & Analytics:

- Time intelligence (YTD, Quarterly)
- Median calculation
- Profit margin computation
- Currency conversion modeling

Visualization:

- KPI design
- Comparative analysis visuals
- Dashboard storytelling
- Executive reporting structure

Financial Analysis:

- Revenue trend evaluation
- Profitability analysis
- Product performance evaluation
- Regional market assessment

17. Project Reflection

This project reinforced the importance of:

- Clean data as the foundation of reliable analysis.
- Structured modeling for scalable reporting.
- Combining financial analysis with business intelligence tools.
- Designing dashboards that communicate insights clearly to decision-makers.

It demonstrates the ability to bridge technical Power BI skills with financial strategy and business impact.

Conclusion

This project showcases a full end-to-end business intelligence workflow for Tailwind Traders. By cleaning, validating, and enriching the dataset, developing a snowflake-modeled data structure, and creating user-defined tables and DAX measures, the project provides accurate, multi-dimensional insights.

The interactive dashboards highlight KPIs such as stock, median sales, loyalty points, net revenue, profit YTD %, and gross revenue trends. Key business insights include the UK having the highest loyalty points, UAE achieving the greatest median sales, Gardening products as the top-selling category, and the Modular Sofa Set generating the highest revenue.

The project demonstrates strong capabilities in data preparation, modeling, DAX calculations, and dashboard development, providing Tailwind Traders with a robust, scalable tool to monitor performance, optimize profitability, and make informed strategic decisions.